

Databricks Questions 1-36

Complete multiple-choice practice set with all options and the correct answers

Contents

Questions 1-12: ingestion, data quality, governance, and security

Questions 13-24: secrets, optimization, streaming, CI/CD, and collaboration

Questions 25-36: Unity Catalog, orchestration, performance, sharing, and access control

Note: Wording has been lightly normalized from the supplied photographs for readability; answer choices and intended meaning are preserved.

Part I - Questions 1-12

Question 1

While reviewing a query's execution in the Databricks Query Profile, a data engineer observes that the Top operators panel shows a sort operator with high Time spent and Memory peak metrics, and the Spark UI reports frequent data spilling. How should the data engineer address this issue?

- A. Repartition the DataFrame to a single partition before sorting.
- B. Convert the sort operation to a filter operation.
- C. Switch to a broadcast join to reduce memory usage.
- D. Increase the number of shuffle partitions to better distribute data.

Correct answer: D. Increase the number of shuffle partitions to better distribute data.

Question 2

A data engineer is designing a system using Lakeflow Declarative Pipelines to process real-time truck telemetry ingested from JSON files in S3 using Auto Loader. The system requires near-real-time latest location, speed, and fuel level per truck, plus daily aggregated distance and fuel-efficiency reports. Which approach should be used?

- A. Use a streaming table for raw ingestion and daily aggregates, and a materialized view for the latest truck status.
- B. Use a streaming table for raw ingestion, a streaming table to incrementally compute the latest truck status, and a materialized view for the daily aggregates.
- C. Use a streaming table for raw ingestion, a materialized view for the latest truck status, and another materialized view for daily aggregates.
- D. Use a materialized view for raw ingestion, a streaming table for latest truck status, and another materialized view for daily aggregates.

Correct answer: B. Use a streaming table for raw ingestion, a streaming table to incrementally compute the latest truck status, and a materialized view for the daily aggregates.

Question 3

A healthcare analytics team is implementing a dimensional model in Delta Lake for patient-care analysis. Which date-dimension design supports efficient time-based querying and aggregation?

- A. Pre-calculate attributes such as fiscal_period, quarter, month_name, day_of_week, and holiday.
- B. Create separate dimension tables for different calendar systems such as fiscal and academic calendars.
- C. Store only the date value and calculate every time attribute in queries.
- D. Store the date as an ISO-formatted string (YYYY-MM-DD) for readability.

Correct answer: A. Pre-calculate attributes such as fiscal_period, quarter, month_name, day_of_week, and holiday.

Question 4

A security team must restrict sales users to customers in their region and mask email addresses for non-admin users. Which implementation should be used with Unity Catalog row filters and column masks?

- A. Create a view with dynamic WHERE clauses and use string replacement functions for masking through ALTER COLUMN SET MASK.
- B. Implement only row filters because column masks cannot be combined with row filters on the same table, then recreate the table.
- C. Create SQL UDFs for regional row filtering and group-based column masking, then apply them using ALTER TABLE SET ROW FILTER and ALTER COLUMN SET MASK.
- D. Use table ACLs with tags and rely on application-level filtering for the user's region.

Correct answer: C. Create SQL UDFs for regional row filtering and group-based column masking, then apply them using ALTER TABLE SET ROW FILTER and ALTER COLUMN SET MASK.

Question 5

What describes a primary technical challenge in ensuring consistent PII masking across all nodes in large-scale distributed Databricks batch and streaming pipelines?

- A. Native masking automatically synchronizes with all downstream external Databricks systems.
- B. Dynamic data masking is applied only at rest, so it does not affect query performance.
- C. Masking functions must be standardized and managed through Unity Catalog, with enforcement across all relevant datasets to avoid inconsistency.
- D. PII masking is required only for direct identifiers.

Correct answer: C. Masking functions must be standardized and managed through Unity Catalog, with enforcement across all relevant datasets to avoid inconsistency.

Question 6

A data engineer is copying a sample of a production sales table to a sandbox. The only PII field and user identifier is user_email. Which anonymization code should be used?

- A. `df.withColumn('user_email', F.expr('uuid()'))`
- B. `df.withColumn('hashed_email', sha2('user_email'))`
- C. `df.withColumn('user_email', F.regexp_replace('user_email', '@.*', '@anonymized.com'))`
- D. `df.withColumn('user_email', F.sha2('user_email'))`

Correct answer: D. `df.withColumn('user_email', F.sha2('user_email'))`

Question 7

A SQL Alert must trigger only when the percentage of customers with missing email addresses exceeds 15% AND the percentage with invalid phone formats exceeds 10%. Which SQL query pattern is appropriate?

- A. Return COUNT(*) for rows where email is NULL OR the phone format is invalid.
- B. Calculate both percentages in a subquery, then return 1 with CASE only when email_null_pct > 15 AND phone_invalid_pct > 10; otherwise return 0.
- C. Calculate and return both percentages without evaluating the combined threshold condition.
- D. Return customer rows where email is NULL AND the phone format is invalid.

Correct answer: B. Calculate both percentages in a subquery, then return 1 with CASE only when email_null_pct > 15 AND phone_invalid_pct > 10; otherwise return 0.

Question 8

Why are Pandas UDFs often preferred over traditional PySpark UDFs for performance-critical applications involving large datasets?

- A. They eliminate the JVM-Python boundary by bypassing serialization entirely.
- B. They provide row-level Python execution with native Spark optimization and remove the need for columnar execution.
- C. They use Apache Arrow for vectorized operations between JVM and Python runtimes, reducing serialization costs and improving computational efficiency.
- D. They stream each row individually through a lightweight Python wrapper, avoiding batch-processing overhead.

Correct answer: C. They use Apache Arrow for vectorized operations between JVM and Python runtimes, reducing serialization costs and improving computational efficiency.

Question 9

A Delta table filter on event_date reads almost all files despite the predicate event_date > '2024-01-01'. What is the probable reason for poor data skipping?

- A. The event_date column is outside the table's partitioning and Z-ordering scheme.
- B. The filter is executed only after a full data scan, which prevents data skipping.
- C. The Delta table lacks an optimization that enables dynamic file pruning.
- D. The filter uses a data type excluded from data-skipping support.

Correct answer: A. The event_date column is outside the table's partitioning and Z-ordering scheme.

Question 10

The AUTO CDC API must propagate deletions from a CDF-enabled source to a target, but some delete events arrive out of order. How does AUTO CDC ensure deletions are applied correctly?

- A. It runs VACUUM on the target table to purge conflicting records.
- B. It ignores deletions that arrive after updates for the same key.
- C. It uses sequence_by to order events and retains tombstones for deleted rows until older sequences are processed.
- D. It manually sorts all incoming events by timestamp before applying changes.

Correct answer: C. It uses sequence_by to order events and retains tombstones for deleted rows until older sequences are processed.

Question 11

A central IT team provisions a Unity Catalog catalog and wants a project group to create and manage schemas, tables, volumes, and related objects, but not rename, delete, or change permissions on the catalog. Which rights should the group receive?

- A. Make the group OWNER of the catalog.
- B. Grant the group ALL PRIVILEGES on the catalog.
- C. Grant the group ALL PRIVILEGES and MANAGE on the catalog.
- D. Grant only USE CATALOG and USE SCHEMA on the catalog.

Correct answer: B. Grant the group ALL PRIVILEGES on the catalog.

Question 12

A faulty IoT sensor reports -500 C, violating an expectation that permits values from -100 C to 200 C. The engineer wants to analyze the faulty data while preserving data-quality standards. What should be done?

- A. Fix the pipeline and implement quarantine logic to isolate the faulty data before rerunning it.
- B. Change the expectation action from fail to warn so invalid records remain in the production output.
- C. Remove all expectations to prevent future failures.
- D. Ignore the error and rerun because Databricks will automatically skip the bad record.

Correct answer: A. Fix the pipeline and implement quarantine logic to isolate the faulty data before rerunning it.

Part II - Questions 13-24

Question 13

A fraud-detection pipeline calls the OpenAI API and needs an access token stored as a Databricks secret. Which CLI command and retrieval order are correct?

- A. databricks secrets put-secret KEY SCOPE; dbutils.secrets.get(KEY, SCOPE)
- B. databricks tokens put-token KEY SCOPE; dbutils.tokens.get(KEY, SCOPE)
- C. databricks tokens put-token SCOPE KEY; dbutils.tokens.get(SCOPE, KEY)
- D. databricks secrets put-secret SCOPE KEY; dbutils.secrets.get(SCOPE, KEY)

Correct answer: D. databricks secrets put-secret SCOPE KEY; dbutils.secrets.get(SCOPE, KEY)

Question 14

A platform lead must automate individual-user attribution for SQL Warehouse usage and share a daily report with executives. What should be implemented?

- A. Use system tables for audit and billing usage, and share raw queries for executives to run manually.
- B. Use system tables for audit and billing usage, create a dashboard with a daily refresh schedule, and share it with the executive team.
- C. Require users to provide all query details before they can run SQL.
- D. Ask users to manually report warehouse usage to executives.

Correct answer: B. Use system tables for audit and billing usage, create a dashboard with a daily refresh schedule, and share it with the executive team.

Question 15

Predictive Optimization is enabled by default for Unity Catalog managed tables. Which two listed operations does it run to maintain Delta tables? (Choose 2.)

- A. OPTIMIZE
- B. PARTITION BY
- C. COMPACT
- D. BUCKETING
- E. ANALYZE

Correct answer: A. OPTIMIZE | E. ANALYZE

Question 16

A streaming pipeline must incrementally ingest JSON files from cloud storage into Delta, automatically handle schema evolution, provide exactly-once processing, and minimize infrastructure management. Which approach meets these requirements?

- A. Use Lakeflow Spark Declarative Pipelines with a static DataFrame read and enable Delta autoMerge.
- B. Use Auto Loader in batch mode with a daily overwrite job.
- C. Use traditional Structured Streaming with Auto Loader while manually configuring checkpoints and mergeSchema.
- D. Use Lakeflow Spark Declarative Pipelines with Auto Loader and cloudFiles.schemaEvolutionMode='addNewColumns'.

Correct answer: D. Use Lakeflow Spark Declarative Pipelines with Auto Loader and cloudFiles.schemaEvolutionMode='addNewColumns'.

Question 17

A team is implementing an append-only Delta Lake pipeline and must prevent existing data from being modified or deleted. Which feature should be enabled?

- A. Delta Time Travel
- B. Delta APPEND_ONLY
- C. Delta OPTIMIZE
- D. Delta VACUUM

Correct answer: B. Delta APPEND_ONLY

Question 18

A data engineer needs an efficient pipeline that automatically processes new CSV files as they arrive in S3. Which Databricks feature should be used?

- A. Traditional batch processing with scheduled Databricks Jobs.
- B. COPY INTO with parameters to track processed files.
- C. Auto Loader with schema inference and evolution enabled.
- D. Standard Spark readStream configured with both CSV and JSON formats.

Correct answer: C. Auto Loader with schema inference and evolution enabled.

Question 19

A message bus may deliver duplicate messages as much as a week apart, and each record must appear exactly once in a Delta target. Which ingestion pattern handles duplicates that may arrive outside the configured watermark?

- A. Write using MERGE INTO with a unique key.
- B. Use Delta Lake Change Data Feed to filter duplicate records.
- C. Use Delta Lake time travel to identify and remove duplicates.
- D. Use Structured Streaming dropDuplicates only.

Correct answer: A. Write using MERGE INTO with a unique key.

Question 20

A governance team must rapidly generate meaningful comments for thousands of undocumented columns using column names, data types, sample values, and data patterns. Which Databricks approach should it use?

- A. Use DESCRIBE TABLE and manually document columns.
- B. Use custom PySpark with df.describe() and df.schema to generate only statistical descriptions.
- C. Use DESCRIBE HISTORY to infer column purpose from table changes.
- D. In Catalog Explorer, open the table schema and use AI Generate to create column descriptions.

Correct answer: D. In Catalog Explorer, open the table schema and use AI Generate to create column descriptions.

Question 21

A CI/CD pipeline uses Databricks Asset Bundles to deploy and run an integration-test job for a pull request. What is the correct bundle command sequence?

- A. init, deploy, run, validate
- B. validate, deploy, run
- C. deploy, run, validate
- D. init, validate, deploy, run

Correct answer: B. validate, deploy, run

Question 22

A PySpark DataFrame contains customer_id, millisecond-precision transaction_timestamp, and amount_spent. Which window specification correctly computes a cumulative sum per customer from the earliest timestamp through the current row?

- A. Window.partitionBy('customer_id').orderBy('transaction_timestamp') with no explicit frame.
- B. Window.partitionBy('customer_id').orderBy('transaction_timestamp').rowsBetween(Window.unboundedPreceding, Window.currentRow).
- C. Window.partitionBy('customer_id').rowsBetween(Window.unboundedPreceding, Window.currentRow), without orderBy.
- D. Window.orderBy('transaction_timestamp').rowsBetween(Window.unboundedPreceding, Window.currentRow), without partitionBy.

Correct answer: B. Window.partitionBy('customer_id').orderBy('transaction_timestamp').rowsBetween(Window.unboundedPreceding, Window.currentRow).

Question 23

A Databricks team wants each member to develop and test independently, avoid accidental overwrites and branch-switching conflicts, and integrate work through CI/CD. How should they collaborate?

- A. All members use one shared Databricks Git folder and switch branches there.
- B. Each member creates a personal Databricks Git folder mapped to the same remote repository and works on their own development branch.
- C. Members edit notebooks in a shared workspace folder and periodically copy them into Git.
- D. Members clone the repository through a cluster web terminal and perform Git operations there.

Correct answer: B. Each member creates a personal Databricks Git folder mapped to the same remote repository and works on their own development branch.

Question 24

A streaming pipeline must route negative transaction values to quarantine. Good data has low-latency production dependencies; quarantined data is analyzed only periodically. The quarantine process must not affect production and cost should be minimized. How should it be implemented?

- A. Keep all records together and add a quarantined Boolean column.
- B. Use a single foreachBatch to write good and bad data, and wrap writes in try/catch.
- C. Filter bad rows from the production stream, and run a separate quarantine stream on a small cluster only once per day.
- D. Run both the good-data and quarantine streams separately on the same cluster.

Correct answer: C. Filter bad rows from the production stream, and run a separate quarantine stream on a small cluster only once per day.

Part III - Questions 25-36

Question 25

Which SQL command programmatically adds tags to a Unity Catalog table?

- A. ALTER TABLE table_name SET TAGS ('key1' = 'value1', 'key2' = 'value2');
- B. APPLY TAGS ON table_name VALUES ('key1' = 'value1', 'key2' = 'value2');
- C. COMMENT ON TABLE table_name TAGS ('key1' = 'value1', 'key2' = 'value2');
- D. SET TAGS FOR table_name AS ('key1' = 'value1', 'key2' = 'value2');

Correct answer: A. ALTER TABLE table_name SET TAGS ('key1' = 'value1', 'key2' = 'value2');

Question 26

A workspace was automatically enabled for Unity Catalog, creating a workspace catalog. New users can create tables in the default schema but cannot access tables in other schemas in the same catalog. Why?

- A. Workspace-catalog permissions are not subject to inheritance rules.
- B. Workspace users receive USE CATALOG and specific privileges on the default schema only.
- C. Tables in other schemas require additional BROWSE privileges that users never receive automatically.
- D. New users receive CREATE TABLE only on the default schema.

Correct answer: B. Workspace users receive USE CATALOG and specific privileges on the default schema only.

Question 27

A Delta table receives frequent account-level UPDATE corrections. The company wants to avoid rewriting entire Parquet files for each change. Which feature should it enable?

- A. Deletion vectors
- B. Partitioning by account_id
- C. Automatic file compaction on writes
- D. Change Data Feed

Correct answer: A. Deletion vectors

Question 28

A job must download ten changeable report types from a slow, intermittently failing REST API, track each activity, retry only failures, and scale throughput. How should it be designed?

- A. Define ten fixed notebook tasks.
- B. Track ten status rows in Delta and execute downloads through a Pandas UDF.
- C. Use a For Each task with the report-type list as input.
- D. Loop through report types inside one notebook task and print the results.

Correct answer: C. Use a For Each task with the report-type list as input.

Question 29

A `groupBy('region').agg(sum('revenue'))` operation shows excessive shuffling. Which technique should be applied to reduce shuffle during aggregation?

- A. Use a broadcast join.
- B. Cache the source DataFrame.
- C. Use `coalesce(1)` after aggregation.
- D. Repartition by region before aggregation.

Correct answer: D. Repartition by region before aggregation.

Question 30

An external script should run only if a particular Databricks job has completed successfully today. Which databricks jobs list-runs options correctly retrieve today's completed executions?

- A. `--job-id JOB_ID --start-time-to TODAY_MIDNIGHT_EPOCH_MS --active-only`
- B. `--job-id JOB_ID --start-time-from TODAY_MIDNIGHT_EPOCH_MS --active-only`
- C. `--job-id JOB_ID --start-time-to TODAY_MIDNIGHT_EPOCH_MS --completed-only`
- D. `--job-id JOB_ID --start-time-from TODAY_MIDNIGHT_EPOCH_MS --completed-only`

Correct answer: D. `--job-id JOB_ID --start-time-from TODAY_MIDNIGHT_EPOCH_MS --completed-only`

Question 31

A company must securely share customer analytics with Partner A, which uses Databricks with Unity Catalog, and Partner B, which uses Apache Spark on AWS without Databricks. Which approach is correct?

- A. Use Databricks-to-Databricks sharing for Partner A; use the open Delta Sharing protocol with bearer tokens or OIDC federation for Partner B.
- B. Use open sharing and bearer tokens for both partners.
- C. Use open sharing for both because it is always more secure than Databricks-to-Databricks sharing.
- D. Databricks-to-Databricks sharing works only within the same cloud, and Unity Catalog governance cannot apply to external sharing.

Correct answer: A. Use Databricks-to-Databricks sharing for Partner A; use the open Delta Sharing protocol with bearer tokens or OIDC federation for Partner B.

Question 32

Auto Loader reads JSON with schema 'a int, b int' and writes invalid records to badRecordsPath. Some well-formed JSON records are quarantined. What is the cause?

- A. The engineer forgot cloudFiles.quarantineMode='rescue'.
- B. The upstream provider switched all data to multiline JSON.
- C. The JSON is syntactically valid but does not conform to the defined schema.
- D. The badRecordsPath contains too many small files.

Correct answer: C. The JSON is syntactically valid but does not conform to the defined schema.

Question 33

A senior data engineer is reviewing claims about Delta Lake while designing scalable data models. Which recommendation is not valid and can be ignored?

- A. Delta Lake optimizes metadata handling and can scale to very large datasets and file counts.
- B. Delta Lake works with formats such as Parquet, JSON, and CSV and integrates with Spark and Databricks tools.
- C. Delta Lake supports both batch and streaming processing.
- D. Delta Lake provides limited monitoring and troubleshooting support, so partner tools must be identified and configured.

Correct answer: D. Delta Lake provides limited monitoring and troubleshooting support, so partner tools must be identified and configured.

Question 34

When monitoring a complex workload, where can the query-plan visualization be found?

- A. In the Spark UI, under the SQL/DataFrame tab.
- B. In Query Profiler, under the Stages tab.
- C. In Query Profiler, under Query Source.
- D. In the Spark UI, under the Jobs tab.

Correct answer: A. In the Spark UI, under the SQL/DataFrame tab.

Question 35

A data engineer is optimizing MERGE on an 800 GB Unity Catalog managed table with frequent updates and deletions. Which two actions should be prioritized? (Choose 2.)

- A. Overwrite the table instead of using MERGE.
- B. Partition the table by date.
- C. Apply liquid clustering using the MERGE join keys.
- D. Enable deletion vectors if they are not already enabled.
- E. Use ZORDER on high-cardinality columns.

Correct answer: C. Apply liquid clustering using the MERGE join keys. | D. Enable deletion vectors if they are not already enabled.

Question 36

A Unity Catalog table contains sensitive fields such as SSN and credit score. Interns should see masked values, while analysts should access only rows for their assigned region. How should the policy be enforced without duplicating data?

- A.** Create views using `current_user()` and `is_account_group_member()`, with masking logic repeated in each `SELECT` clause.
- B.** Use Unity Catalog row filters based on region and column masks based on user roles.
- C.** Create dynamic views for each user role and manage them with ACLs.
- D.** Use row filters based on user roles and column masks based on region.

Correct answer: B. Use Unity Catalog row filters based on region and column masks based on user roles.